

Bharani Shree R

Python Developer

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Summary

Python Developer with 2 years of experience building scalable backend systems using FastAPI and Django. Skilled in designing secure REST APIs with JWT/RBAC, optimizing PostgreSQL and MySQL databases, and implementing asynchronous workflows with Celery and Redis. Experience integrating AI-assisted API execution using the Model Context Protocol (MCP) with Claude. Passionate about clean architecture, performance optimization, and Agile development practices.

Experience

B2E Technology, Chennai — Python Developer

June 2024 – May 2026

FastAPI | Django | DRF | JWT | RBAC | Celery | Redis | PostgreSQL | MySQL | SQLAlchemy | Docker | GitHub Actions

- Developed 50+ RESTful APIs using FastAPI and Django with JWT authentication and Role-Based Access Control (RBAC).
- Built responsive ERP dashboards using HTML, CSS, Bootstrap, and JavaScript, integrated with backend REST APIs.
- Optimized PostgreSQL and MySQL queries and schemas, improving performance across ERP platforms.
- Implemented asynchronous background task processing using Celery and Redis for scheduled and event-driven workflows.
- Containerized applications using Docker and automated deployment pipelines with GitHub Actions.
- Integrated third-party APIs and SMTP services, and wrote Pytest unit and integration tests.
- Actively participated in Agile/Scrum development cycles and peer code reviews.

Technical Skills

Languages: Python, SQL, JavaScript

Frameworks: Django, FastAPI, Django REST Framework (DRF), SQLAlchemy, Pydantic

Databases: PostgreSQL, MySQL, SQLite, Redis

Cloud: AWS (EC2, S3, IAM)

Tools: Docker, Git, GitHub Actions, Postman, Swagger/OpenAPI, Pytest, DBeaver, VS Code, PyCharm

Core Skills: REST APIs, JWT/OAuth2, RBAC, Celery, Async Processing, OOP, Microservices, CI/CD Pipelines, MCP (Model Context Protocol), Performance Optimization, Agile/Scrum

Projects

ERP Management System

FastAPI | MySQL | SQLAlchemy | JWT

- Designed backend services using FastAPI and DRF for staff, student, and academic workflows, including approval chains and condition-based execution flows.
- Improved system performance through query indexing and reduced data processing time via API-level optimizations.
- Implemented JWT and RBAC; documented all endpoints using Swagger/OpenAPI for faster frontend team onboarding.
- Extended the platform with a multi-tenant MIS reporting module (Bootstrap, JavaScript), enabling dynamic department- and academic-year-based report generation for HODs, faculty, and compliance teams.

AI-Integrated API Service using FastAPI and MCP

FastAPI | MCP | Claude | REST API

- Developed a RESTful API using FastAPI for structured data retrieval and processing.
- Integrated the API with an MCP (Model Context Protocol) server, enabling Claude to discover and invoke backend APIs through natural language.
- Defined tool schemas and request/response models to support secure AI-assisted API execution with live responses.

TN Government – Public Health Training Portal

Django | DRF | PostgreSQL | JWT | JavaScript

- Built a centralized government platform managing internship and training workflows across Tamil Nadu's public health department for multiple institutions.
- Designed multi-stage approval workflows to process and track training requests, reducing manual coordination across departments.
- Implemented reporting-friendly PostgreSQL structures for real-time monitoring; enforced RBAC and JWT security; documented APIs via Swagger/OpenAPI.

Education

Master of Computer Applications (MCA) — University of Madras, Chennai

B.Sc. Computer Science — Justice Basheer Ahmed Sayeed College for Women, Chennai